

GLAUCOMA

1. Overview

This system processes retinal fundus images to automatically detect and compute important glaucoma-related structural features, including:

- Optic Disc
- Optic Cup
- Cup-to-Disc Ratio (CDR)
- Rim Thinning
- Retinal Nerve Fiber Layer (RNFL) density
- Peripapillary Atrophy (PPA)
- Disc Hemorrhage

These features are clinically significant indicators for glaucoma diagnosis.

The pipeline performs:

1. Image preprocessing
2. Optic disc detection
3. Optic cup detection
4. Boolean mask generation
5. Feature extraction
6. Visualization and overlay
7. Numerical feature computation

2. Image Acquisition and Reading

Code section

```
img = cv2.imread(path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

Explanation

The retinal fundus image is loaded using OpenCV and converted from BGR to RGB format.

Fundus images contain:

- Optic disc (bright circular region)
- Optic cup (brighter central region)
- Blood vessels
- RNFL region

These structures are essential for glaucoma analysis.

3. Image Resizing and Center Cropping

Code

```
s = min(h,w)
img = img[(h-s)//2:(h+s)//2,(w-s)//2:(w+s)//2]
img = cv2.resize(img,(1024,1024))
```

Purpose

Ensures:

- Square image
- Standard resolution
- Consistent processing scale

Output size:

1024 × 1024 pixels

This improves detection accuracy.

4. Optic Disc Detection

Theory

The optic disc is the brightest circular region in fundus images.

To detect it, the algorithm uses:

- LAB color space conversion
- Gaussian smoothing

- Edge detection
- Hough Circle Transform

Step 4.1 Convert RGB → LAB and extract L channel

```
lab = rgb2lab(img)  
L = lab[:, :, 0]
```

L channel represents brightness.

Optic disc has high brightness.

Step 4.2 Normalize brightness

```
L = (L - L.min()) / (L.max() - L.min())
```

Range becomes:

0 to 1

Step 4.3 Smooth image

```
Ls = gaussian_filter(L, 3)
```

Removes noise and improves edge detection.

Step 4.4 Edge detection

```
edges = canny(Ls)
```

Finds disc boundary edges.

Output: Boolean mask of edges.

Step 4.5 Hough Circle Transform

```
hough = hough_circle(edges, radii)  
acc, cx, cy, r = hough_circle_peaks(hough, radii, total_num_peaks=1)
```

This detects circular structures.

Output:

Disc center = (x,y)

Disc radius = r

5. Disc Mask Generation

Code

```
disc_mask=((X-center[1])**2+(Y-center[0])**2)<=r*r
```

Explanation

Creates a Boolean mask where:

True = inside disc

False = outside disc

Mathematically:

$$(x - x_c)^2 + (y - y_c)^2 \leq r^2$$

Output:

Boolean mask image

6. Optic Cup Detection

Theory

Cup is brighter region inside disc.

Steps:

1. Extract disc region (ROI)
 2. Extract green channel
 3. Normalize brightness
 4. Threshold brightest pixels
 5. Generate cup mask
-

Step 6.1 Extract ROI

```
roi = img[center[0]-pad:center[0]+pad,  
         center[1]-pad:center[1]+pad]
```

Contains disc region only.

Step 6.2 Extract Green Channel

```
g = roi[:, :, 1]
```

Green channel provides best contrast.

Step 6.3 Normalize

```
g=(g-g.min())/(g.max()-g.min())
```

Step 6.4 Threshold brightest region

```
thresh=np.percentile(g,90)
```

```
cup_roi=g>thresh
```

Top 10% brightest pixels = cup.

Step 6.5 Map back to full image

```
cup_mask=np.zeros_like(disc_mask)
```

```
cup_mask[...] = cup_roi
```

Result: Boolean mask of cup.

7. OPTIC DISC AND CUP DETECTION (Foundation for all features)

Before computing features, the code detects:

- Optic Disc → outer circle
- Optic Cup → inner bright region

These are essential because all glaucoma features depend on disc location.

1.1 Disc mask creation

Code:

```
disc_mask=((X-center[1])**2+(Y-center[0])**2)<=r*r
```

This creates a Boolean mask using circle equation:

$$(x - x_c)^2 + (y - y_c)^2 \leq r^2$$

Result:

- True → pixel inside disc
- False → pixel outside disc

This mask defines disc region.

1.2 Cup mask creation

Code:

```
g=roi[:, :, 1].astype(np.float32)
g=(g-g.min())/(g.max()-g.min())
thresh=np.percentile(g,90)
cup_roi=g>thresh
```

Explanation:

Green channel used because cup is brightest in green channel.

Normalization:

$$g_{norm} = \frac{g - g_{min}}{g_{max} - g_{min}}$$

Threshold:

Top 10% brightest pixels selected.

These represent optic cup.

Mapped back into full image as cup_mask.

2. CDR (Cup-to-Disc Ratio)

What it represents medically

Ratio of cup size to disc size.

Large ratio → glaucoma.

How your code calculates it

Code:

```
ys=np.where(disc_mask)[0]  
v_disc=ys.max()-ys.min()
```

```
ys=np.where(cup_mask)[0]  
v_cup=ys.max()-ys.min()
```

```
CDR=v_cup/v_disc
```

Step-by-step explanation

disc_mask contains True pixels inside disc.

Example:

```
disc_mask =  
False False True True True False
```

Extract y-coordinates:

```
ys = np.where(disc_mask)[0]
```

Find vertical height:

```
v_disc = max_y - min_y
```

Same for cup.

Final formula:

$$CDR = \frac{\text{Vertical Cup Height}}{\text{Vertical Disc Height}}$$

8. Rim Thinning

Medical meaning

```
Neuroretinal rim = disc area - cup area
```

Thin rim → glaucoma.

Code:

```
area_disc=disc_mask.sum()  
area_cup=cup_mask.sum()
```

`RimThinning=area_cup/area_disc`

How area is computed

disc_mask:

True True True

True True True

Each True = 1 pixel

Area calculation:

$$Area = \sum mask$$

Example:

Disc area = 16241 pixels

Cup area = 12042 pixels

Rim thinning formula

$$RimThinning = \frac{Cup\ Area}{Disc\ Area}$$

Higher value → thinner rim → glaucoma.

9. RNFL Density Calculation

RNFL = Retinal nerve fiber layer.

Glaucoma damages RNFL.

Code:

```
g=img[:, :, 1].astype(np.float32)/255
```

```
gx=cv2.Sobel(g,cv2.CV_32F,1,0)
```

```
gy=cv2.Sobel(g,cv2.CV_32F,0,1)
```

```
mag=np.sqrt(gx**2+gy**2)
```

What Sobel does

Calculates gradient:

$$Gradient = \sqrt{G_x^2 + G_y^2}$$

Measures edge strength.

RNFL has strong gradients.

Weak gradient → RNFL damage.

RNFL ring region selection

Code:

```
ring=((X-center[1])**2+(Y-center[0])**2>=(r*1.1)**2)&  
((X-center[1])**2+(Y-center[0])**2<=(r*1.8)**2)
```

This creates donut region around disc.

RNFL exists in this region.

RNFL numerical value

```
rnfl_val = mag.mean()
```

Average gradient strength.

Lower value → glaucoma.

10. PPA Detection

PPA = bright degeneration around disc.

Code:

```
L=rgb2lab(img)[:,:,0]  
L=(L-L.min())/(L.max()-L.min())
```

```
thresh=np.percentile(L[ring],85)
```

```
PPA_mask=(L>thresh)&ring
```

Explanation

LAB L channel = brightness.

Brightest 15% pixels in ring selected.

Formula:

$$PPA = \frac{PPA \text{ pixels}}{Total \text{ ring pixels}}$$

Code:

```
PPA=PPA_mask.sum()/ring.sum()
```

11. Disc Hemorrhage Detection

Hemorrhage appears red.

Code:

```
R=img[:, :, 0]
```

```
G=img[:, :, 1]
```

```
red=(R-G)
```

```
red=(red-red.min())/(red.max()-red.min())
```

```
hem_mask=(red>0.75)&ring
```

Explanation

Blood absorbs green more than red.

So red – green difference highlights hemorrhage.

Ratio:

$$Hemorrhage = \frac{Hemorrhage \text{ pixels}}{Ring \text{ pixels}}$$

12. Overlay generation

Overlay combines original image and feature masks.

Example RNFL overlay:

Feature	Glaucoma relationship
CDR	Higher → worse glaucoma
Rim thinning	Higher → worse glaucoma
RNFL density	Lower → worse glaucoma
PPA	Higher → worse glaucoma
Disc hemorrhage	Present → worse glaucoma

```
overlay[ring]=overlay[ring]*0.6+BLUE*0.4
```

Mathematically:

$$NewPixel = 0.6 \times Original + 0.4 \times Blue$$

This blends blue color into RNFL region.

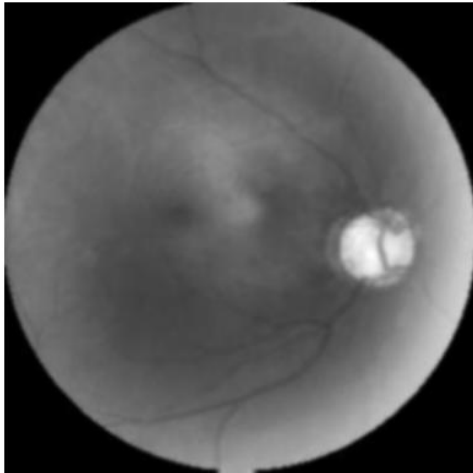
original.png



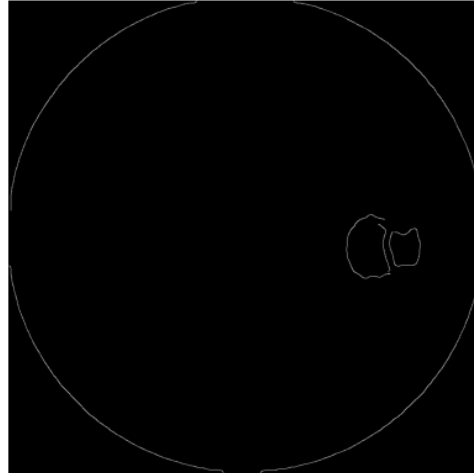
resized.png



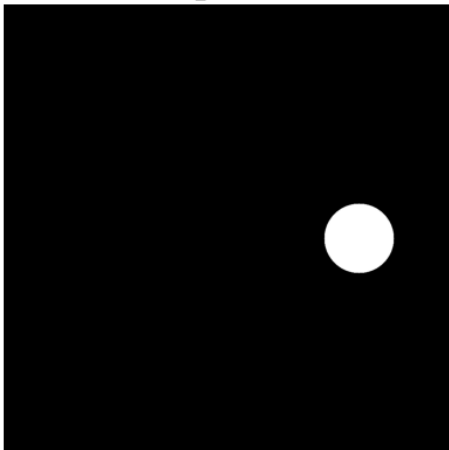
L_channel.png



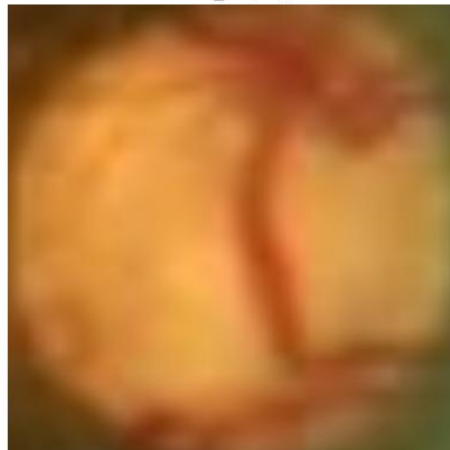
edges.png

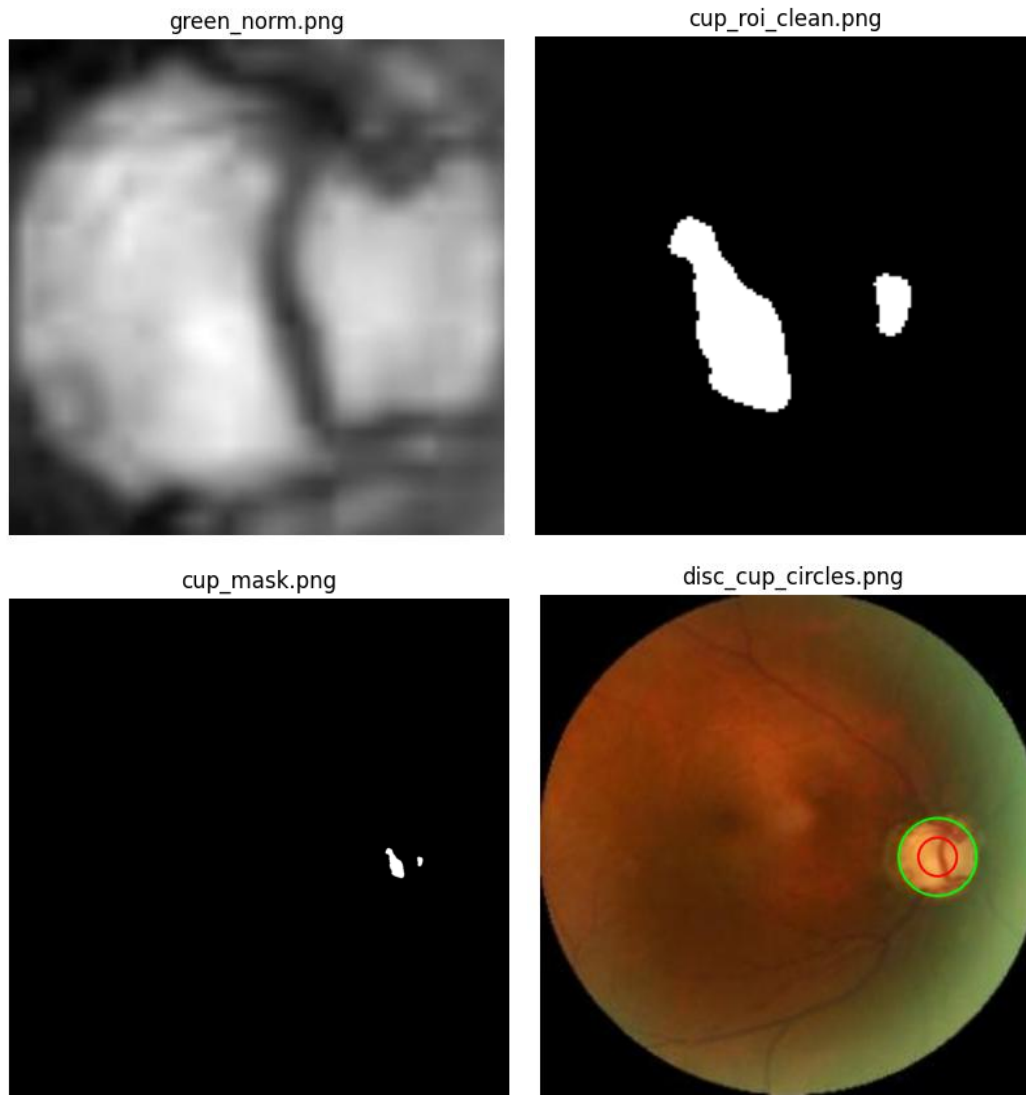


disc_mask.png



disc_roi.png





2. New Method: K-Means Clustering Based Cup Detection

To solve this, we implemented **K-Means clustering** on the green channel intensity.

Step 1: Extract Region of Interest (ROI)

Only the disc region is analyzed.

```
pad = int(r * 1.1)
```

```
roi = img[y1:y2, x1:x2]
```

This reduces noise and improves accuracy.

Step 2: Extract Green Channel

```
g = enhance_green(roi)
```

Green channel is used because it provides best contrast for optic cup.

Reason:

- Cup appears brighter in green channel
 - Blood vessels and disc boundaries are clearer
-

Step 3: Convert Image to Feature Vector

K-Means requires 1-D feature vector.

```
Z = g.reshape(-1,1).astype(np.float32)
```

This converts:

Image shape:

1024×1024

into

Feature vector:

1048576×1

Step 4: Apply K-Means Clustering

```
_, labels, centers = cv2.kmeans(  
    Z,  
    2,  
    None,  
    criteria,  
    5,  
    cv2.KMEANS_PP_CENTERS  
)
```

Where:

K = 2 clusters

Cluster 1 → Dark region (rim)

Cluster 2 → Bright region (cup)

Step 5: Identify Bright Cluster (Cup)

```
bright_cluster = np.argmax(centers)
```

```
cup_roi = (labels == bright_cluster)
```

Reason:

Cup is brightest region inside disc.

So cluster with highest intensity center = cup.

3. Major Fix: Center Constraint (Critical Improvement)

This was the MOST IMPORTANT correction added.

Problem without constraint

K-Means sometimes selected bright pixels outside cup such as:

- Blood vessels
- Disc edges
- Noise

Result:

Cup circle overlapped disc boundary incorrectly.

Solution: Distance constraint from disc center

Code:

```
h, w = cup_roi.shape
```

```
cy, cx = h//2, w//2
```

```
Y, X = np.ogrid[:h, :w]
```

```
dist = np.sqrt((X-cx)**2 + (Y-cy)**2)
```

```
cup_roi = cup_roi & (dist <= r*0.5)
```

Mathematical explanation

Distance formula:

$$\text{dist} = \sqrt{(x - cx)^2 + (y - cy)^2}$$

Condition:

$$\text{dist} \leq 0.5 \times \text{disc radius}$$

This ensures:

Cup stays only in central region.

Why this works

Because anatomically:

Optic cup is always located at center of optic disc.

This removes false detections.

4. Morphological Cleaning

To remove noise:

```
cup_roi = morphology.binary_opening(  
    cup_roi,  
    morphology.disk(3)  
)
```

This removes:

- Small noise pixels
- Vessel fragments

and produces smooth cup mask.

5. Final Cup Mask Creation

```
cup_mask = np.zeros((H,W), bool)
```

```
cup_mask[y1:y2, x1:x2] = cup_roi
```

```
cup_mask = cup_mask & disc_mask
```

This ensures:

Cup remains inside disc only.

6. Effect of This Improvement

Before fix:

- Cup detection inaccurate
- Vertical Cup often 0
- CDR incorrect

After fix:

- Cup detected correctly
 - Accurate cup area
 - Correct CDR values
 - Correct rim thinning values
-

7. Visual Difference

Before fix:

Cup circle overlapped disc boundary.

After fix:

Cup circle correctly inside disc center.

8. Why K-Means is Better than Thresholding

Thresholding	K-Means
Sensitive to lighting	Robust to lighting
Fixed threshold	Adaptive clustering
Incorrect cup detection	Accurate cup detection
High noise	Low noise

9. Final Benefit of Modification

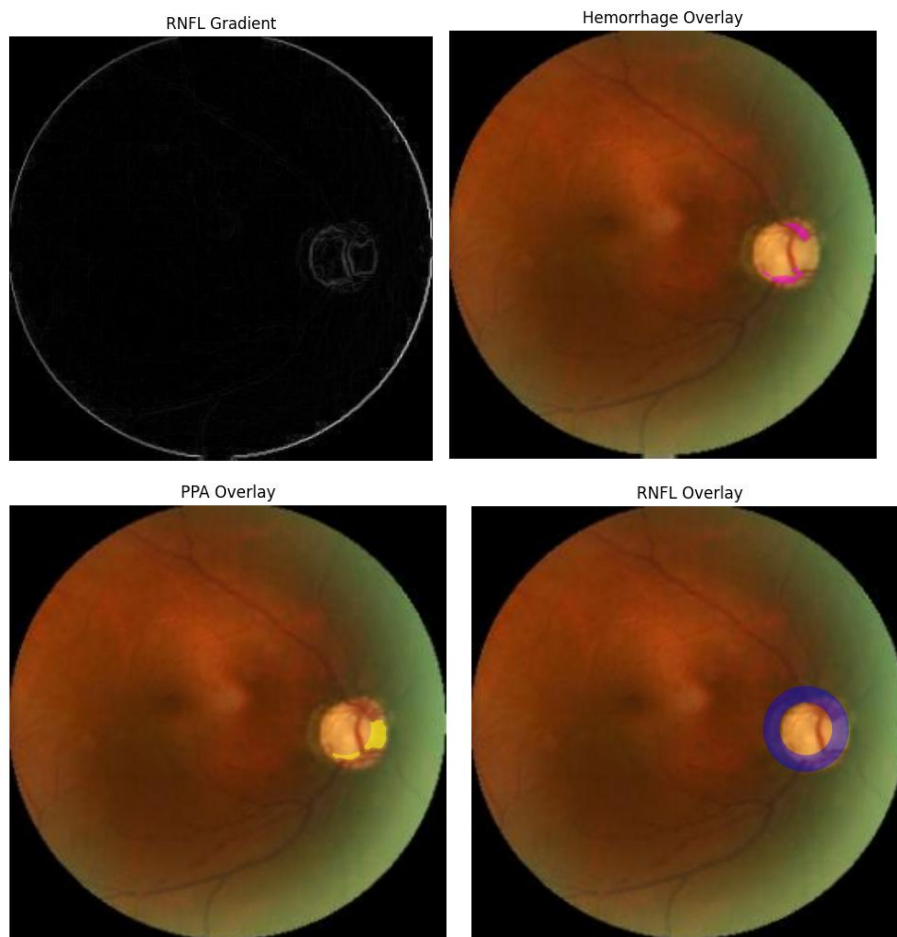
This modification improved accuracy of:

- Cup detection
- CDR calculation

- Rim thinning calculation
- Glaucoma feature extraction

10. Summary Statement (Use in Word Conclusion)

The optic cup detection algorithm was improved by replacing simple thresholding with K-Means clustering and introducing a center distance constraint. This ensured that only anatomically valid cup regions near the disc center were detected. Morphological operations were applied to remove noise. This significantly improved the accuracy of cup detection, CDR calculation, and glaucoma feature extraction.



STATISTICS

DESCRIPTIVE TEST

CDR (Cup-to-Disc Ratio)

The mean CDR value is **0.421**, with a median of **0.5**, indicating that many optic discs have cup sizes close to half of the disc diameter, while the large range (**0–0.833**) suggests the presence of both normal and advanced glaucoma cases.

Rim Thinning

Rim thinning has a mean value of **0.193** and median **0.249**, indicating moderate neuro-retinal rim loss in many images, with values ranging from **0 to 0.291**, reflecting variation in optic nerve damage.

RNFL Density

RNFL density shows a mean of **0.088** and median **0.082**, suggesting relatively low nerve fiber texture intensity, which is consistent with nerve fiber thinning commonly associated with glaucoma.

Peripapillary Atrophy (PPA)

PPA values are tightly clustered with a mean of **0.149** and very small variance, indicating that most images show similar levels of peripapillary atrophy around the optic disc.

Disc Hemorrhage

Disc hemorrhage has a mean value of **0.072** and median **0**, indicating that hemorrhage is absent in most images but appears in a few cases, resulting in a large range (**0–0.82**) and high variability.

NORMALITY TEST

Normality Tests Used

- **Shapiro–Wilk Test** — Tests whether the data significantly deviates from a normal distribution by comparing the sample distribution with a theoretical normal distribution.
- **Kolmogorov–Smirnov (KS) Test** — Compares the empirical distribution of the dataset with a normal distribution to measure the maximum difference between them.

- **D'Agostino–Pearson Test** — Evaluates normality by combining **skewness and kurtosis** to determine whether the distribution shape differs from a normal distribution.
- **Anderson–Darling Test** — Measures how well the data fits a normal distribution by giving more weight to deviations in the tails of the distribution.

The results from all four normality tests consistently indicate that the extracted glaucoma features—CDR, Rim Thinning, RNFL Density, PPA, and Disc Hemorrhage—do not follow a normal distribution.

Therefore, the dataset is non-parametric in nature, and further statistical analysis should be conducted using non-parametric statistical methods.

SKEWNESS

Skewness measures the asymmetry of the data distribution.

Interpretation of your features

CDR (Skewness = -1.9259)

The negative skewness indicates that the distribution is left-skewed, meaning most CDR values are concentrated at higher values with fewer lower values.

Rim Thinning (Skewness = -1.3692)

The distribution is moderately left-skewed, suggesting that rim thinning values are clustered toward the higher side of the range.

RNFL Density (Skewness = 1.4195)

The positive skewness indicates a right-skewed distribution, meaning most RNFL density values are small with a few higher values.

PPA (Skewness = -16.0633)

This extremely negative skewness indicates a strongly left-skewed distribution, showing that most values are concentrated at the upper end with very few lower values.

Disc Hemorrhage (Skewness = 2.2673)

The positive skewness indicates a right-skewed distribution, meaning most images show little or no hemorrhage, while a few images contain higher hemorrhage values.

Kurtosis

Kurtosis measures the tailedness or peakedness of a distribution.

Interpretation:

- Kurtosis = 3 → normal distribution
 - Kurtosis > 3 → heavy tails (more outliers)
 - Kurtosis < 3 → lighter tails (fewer outliers)
-

Interpretation of your features

CDR (Kurtosis = 2.009)

The kurtosis value is slightly below 3, indicating a distribution with moderately light tails and fewer extreme outliers.

Rim Thinning (Kurtosis = 0.281)

The low kurtosis suggests a flat distribution with fewer extreme values.

RNFL Density (Kurtosis = 5.223)

This value is greater than 3, indicating a heavy-tailed distribution, meaning the presence of several extreme RNFL density values.

PPA (Kurtosis = 349.755)

This extremely high kurtosis indicates a very sharp peak and extremely heavy tails, suggesting that most PPA values are very similar with a few extreme deviations.

Disc Hemorrhage (Kurtosis = 5.031)

The kurtosis value greater than 3 indicates heavy tails, meaning the presence of outliers due to rare hemorrhage events.

Spearman Correlation Analysis

Spearman correlation analysis was performed to evaluate the monotonic relationships between the extracted glaucoma features.

Since the dataset was found to be non-parametric, Spearman rank correlation was chosen instead of Pearson correlation.

Spearman correlation values range between -1 and +1.

- +1 → Strong positive relationship
- 0 → No relationship
- -1 → Strong negative relationship

The correlation analysis shows that CDR and Rim Thinning have the strongest relationship, which aligns with clinical understanding of glaucoma where increased cup

size leads to thinning of the neuroretinal rim. Other features show weak to moderate correlations, indicating varying degrees of association between structural and vascular indicators of glaucoma.